

```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.20.0)
Requirement already satisfied: numpy<=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.11.10)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging<=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (2.41.4)
Requirement already satisfied: typing-extensions<=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (4.14.1)
Requirement already satisfied: typing-inspection<=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2.3.0)
Requirement already satisfied: certifi<=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2017.4.17)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4) (0.0.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (0.19.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (7.0.5)
Requirement already satisfied: MarkupSafe<=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2) (2.0.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1) (1.17.0)
Collecting en-core-web-sm==3.8.0
```

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl 12.8/12.8 MB 87.9 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
import nltk
import spacy
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
from collections import Counter
```

```
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
True
```

```
medical_text = """
Diabetes is a chronic disease that affects how the body processes blood sugar.
If untreated, diabetes may cause heart disease, kidney failure, nerve damage and vision problems.
Early diagnosis and proper treatment help improve patient outcomes.
"""
print(medical_text)
```

Diabetes is a chronic disease that affects how the body processes blood sugar.
 If untreated, diabetes may cause heart disease, kidney failure, nerve damage and vision problems.
 Early diagnosis and proper treatment help improve patient outcomes.

```
sentences_nltk = sent_tokenize(medical_text)
words_nltk = word_tokenize(medical_text)
```

```
print("Sentences (NLTK):", sentences_nltk)
print("Words (NLTK):", words_nltk)
```

```
-----
LookupError                                Traceback (most recent call last)
/tmp/ipython-input-3323891974.py in <cell line: 0>()
----> 1 sentences_nltk = sent_tokenize(medical_text)
      2 words_nltk = word_tokenize(medical_text)
      3
      4 print("Sentences (NLTK):", sentences_nltk)
      5 print("Words (NLTK):", words_nltk)

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/nltk/data.py in find(resource_name, paths)
    577     sep = "*" * 70
    578     resource_not_found = f"\n{sep}\n{msg}\n{sep}\n"
--> 579     raise LookupError(resource_not_found)
    580
    581
```

```
LookupError:
*****
Resource punkt_tab not found.
Please use the NLTK Downloader to obtain the resource:

>>> import nltk
>>> nltk.download('punkt_tab')

For more information see: https://www.nltk.org/data.html

Attempted to load tokenizers/punkt_tab/english/

Searched in:
- '/root/nltk_data'
- '/usr/nltk_data'
- '/usr/share/nltk_data'
- '/usr/lib/nltk_data'
- '/usr/share/nltk_data'
- '/usr/local/share/nltk_data'
- '/usr/lib/nltk_data'
- '/usr/local/lib/nltk_data'
*****
```

Next steps: [Explain error](#)

```
import nltk
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
True
```

```
sentences_nltk = sent_tokenize(medical_text)
words_nltk = word_tokenize(medical_text)
```

```
print("Sentences (NLTK):", sentences_nltk)
print("Words (NLTK):", words_nltk)
```

```
Sentences (NLTK): ['\nDiabetes is a chronic disease that affects how the body processes blood sugar.', 'If untreated, diabetes m
Words (NLTK): ['Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the', 'body', 'processes', 'blood', 'suga
```

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(medical_text)
```

```
sentences_spacy = [sent.text for sent in doc.sents]
words_spacy = [token.text for token in doc]
```

```
print("Sentences (spaCy):", sentences_spacy)
print("Words (spaCy):", words_spacy)
```

```
Sentences (spaCy): ['\nDiabetes is a chronic disease that affects how the body processes blood sugar.\n', 'If untreated, diabete
Words (spaCy): ['\n', 'Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the', 'body', 'processes', 'blood'
```

```
stemmer = PorterStemmer()
stems = [stemmer.stem(word.lower()) for word in words_nltk if word.isalpha()]
print("Stemmed Words:", stems)
```

```
Stemmed Words: ['diabet', 'is', 'a', 'chronic', 'diseas', 'that', 'affect', 'how', 'the', 'bodi', 'process', 'blood', 'sugar', '
```

```
lemmatizer = WordNetLemmatizer()
lemmas_nltk = [lemmatizer.lemmatize(word.lower()) for word in words_nltk if word.isalpha()]
print("Lemmas (NLTK):", lemmas_nltk)
```

```
Lemmas (NLTK): ['diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar'
```

```
lemmas_spacy = [token.lemma_ for token in doc if token.is_alpha]
print("Lemmas (spaCy):", lemmas_spacy)
```

```
Lemmas (spaCy): ['Diabetes', 'be', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar'
```

```
print("Original:", words_nltk)
print("Stemmed:", stems)
print("NLTK Lemmas:", lemmas_nltk)
print("spaCy Lemmas:", lemmas_spacy)
```

```
Original: ['Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the', 'body', 'processes', 'blood', 'sugar',
Stemmed: ['diabet', 'is', 'a', 'chronic', 'diseas', 'that', 'affect', 'how', 'the', 'bodi', 'process', 'blood', 'sugar', 'if', '
NLTK Lemmas: ['diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar',
spaCy Lemmas: ['Diabetes', 'be', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar',
```